

Soldering Station

KSGER T12 V3.1

User Manual



This is an unofficial user manual that is not affiliated with KSGER and is not supported or endorsed by KSGER.

This manual is provided “**as is**”, without any guarantee of the completeness or accuracy of the information it contains. The author accepts no responsibility for errors, omissions, damage, injury, or any other consequences resulting from the use of the information contained in this manual.

Copyright © 2026 irpmidav.
Distributed under the license [CC BY-NC 4.0](#).

Rev. 1.0 · 2026-10-08

Contents

1. Station Features	4
2. Conventions	5
3. First Power-On	6
4. Selecting the Interface Language	6
5. Controls.....	7
6. Operating Mode	8
7. Setting the Temperature and User Profiles	9
8. Standby, Sleep and Screen Saver Modes	11
9. Boost Mode	13
10. Tip Selection and Calibration.....	14
11. Station Menu	17
12. Quick Control Reference	24

1. Station Features

The soldering station KSGER T12 is built around an STM32 microcontroller and has a 1.3" OLED display. It is compatible with T12-series soldering tips. Main features and capabilities:

- Heat-up to operating temperature in about 8 seconds;
- Operating temperature range from 50 to 480 °C with a soldering iron power of up to 75 W;
- The T12 soldering tip can be replaced without waiting for it to cool down completely;
- Automatic cold junction compensation at a handle temperature of 9...99 °C;
- Up to five user temperature profiles can be stored;
- Calibration of the tip temperature curve and storage of calibration data for the standard T12 tip types and six user tips;
- Manual or automatic transition to Standby mode after a set time (1...60 min);
- Automatic transition to Sleep mode and to screen saver mode after a set time (1...60 min);
- Built-in real-time clock with rate correction;
- Supply voltage monitoring with automatic shutdown;
- Boost mode, which temporarily raises the tip temperature by a set number of degrees for a set time;
- Adjustable temperature step from 1 to 50 °C;
- Selection of the tip type in use for better temperature accuracy;
- The list of available T12 tip types can be edited for quick selection of the tip in use.

2. Conventions

The following conventions are used in this manual:

	Short press of the encoder button (less than 0.5 s)
	Long press of the encoder button (more than 2 s) or holding the button while turning the encoder
	Turn the encoder one position counterclockwise
	Turn the encoder one position clockwise
	Turn the encoder three or more positions counterclockwise
	Turn the encoder three or more positions clockwise
...	Repeat the action until the desired result is achieved, or wait a moment
or	The same goal can be achieved in another way
+	Actions are performed simultaneously

3. First Power-On

Turn on the station with the switch on the **rear panel**. After power-on, the station starts in the mode selected in menu item **[13. Power On]** (see Section **11. Station Menu**).

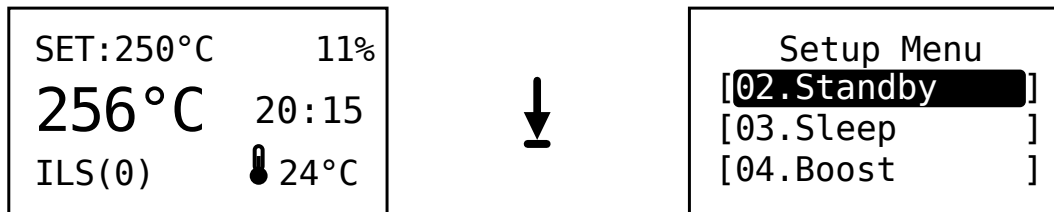
The station can also be switched on without a tip: in this case an error message is shown on the display, but some settings, such as the interface language and the clock, remain available.

During the first power-on with a tip installed, errors may occur during heating, so watch the heating process to make sure the tip does not overheat. If anything looks wrong, switch the station off immediately, let the tip cool down, and check again that the tip is installed correctly.

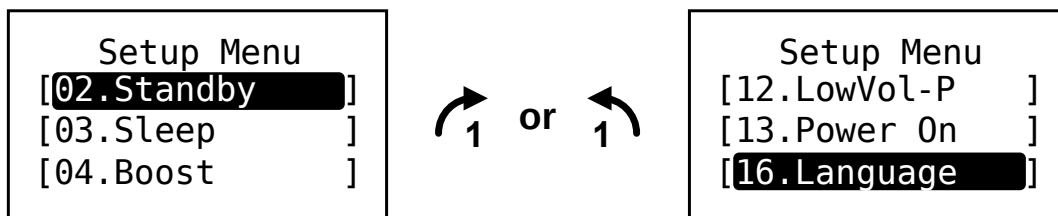
4. Selecting the Interface Language

By default, the interface may be in Chinese. To change the language:

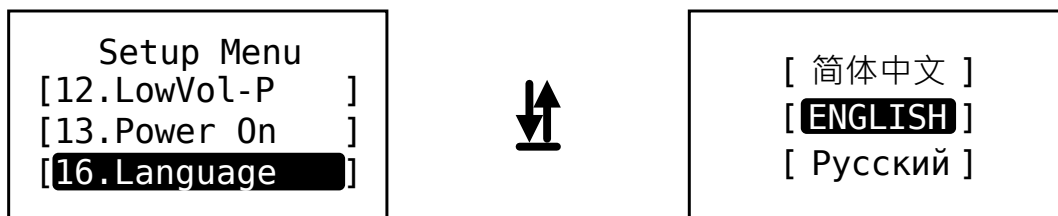
1. Press and hold the encoder button for more than 2 s — this opens the **Setup Menu**:



2. Turn the encoder smoothly in either direction to select the item **[16. Language]**:



3. Press the button briefly — the language selection menu opens:



4. Turn the encoder to select the desired language and press the button briefly — the language is saved and the station switches to operating mode.

5. Controls

All functions of the station are controlled with the rotary encoder, which allows the following actions:

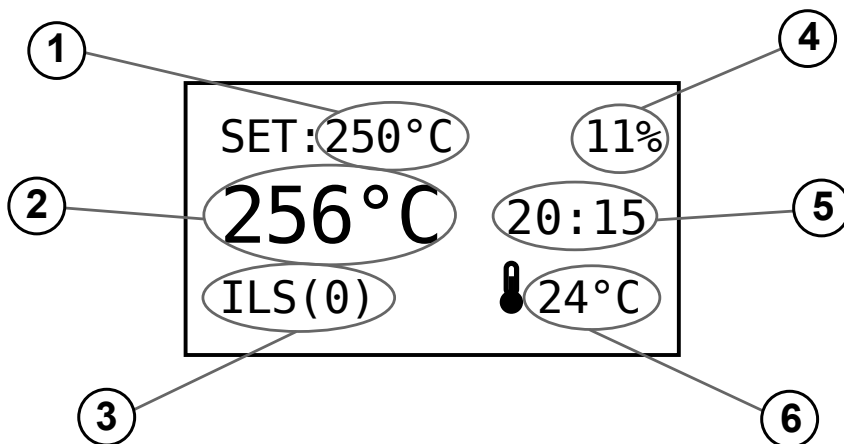
- turning the encoder one position in either direction;
- turning the encoder three or more positions in either direction;
- a short press of the button (less than 0.5 s);

- a long press of the button (more than 2 s);
- turning the encoder while the button is pressed and held.

The functions of these actions in operating mode are summarized in the table in Section 12.

6. Operating Mode

Operating mode is the main mode, in which soldering is performed. In this mode the soldering tip is heated to the operating temperature selected in the current profile (**G1–G5**). In operating mode the display shows the following elements:

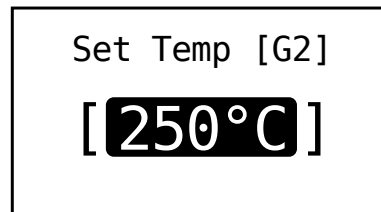
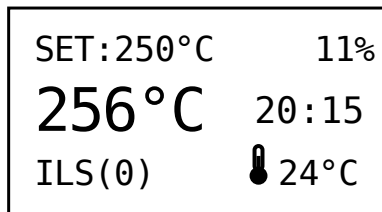


1. Set tip temperature;
2. Current temperature according to the T12 tip sensor;
3. Type of tip in use (selected on the screen **Tip Select**, see Section **10. Tip Selection and Calibration**). If the tip has not been calibrated, **(0)** is shown next to the type; after calibration, **(*)** is shown next to the tip type. Instead of the tip type, the supply voltage of the soldering station can also be shown (see menu **[11. Voltage]**);
4. Power supply load while heating the tip (in percent);
5. Current time from the built-in clock (hours:minutes);
6. Handle temperature from the built-in NTC sensor (if installed). It is approximately equal to room temperature and is used by the station for “cold junction” compensation.

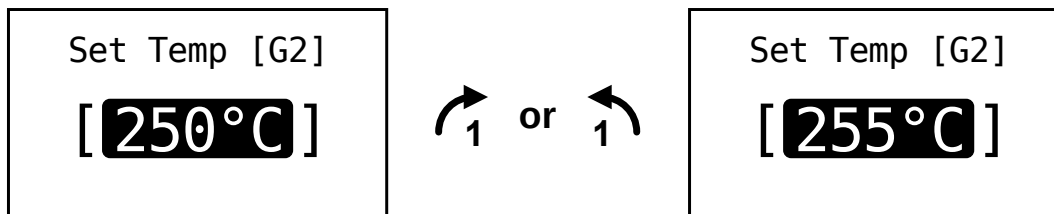
7. Setting the Temperature and User Profiles

The operating temperature of the tip is set using one of five user profiles (**G1–G5**). To change the operating temperature of the selected profile, do the following:

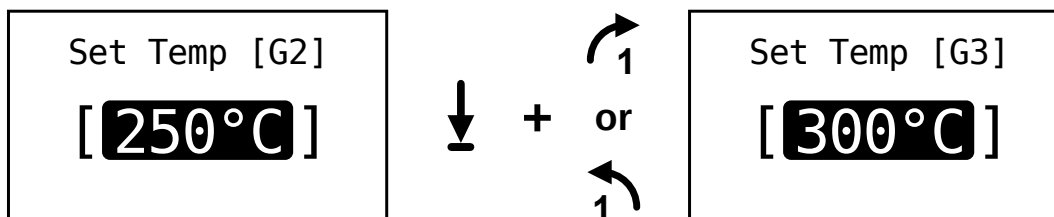
1. In operating mode, press the encoder button briefly — this opens the screen **Set Temp** with the profile number (for example, **[G2]**):



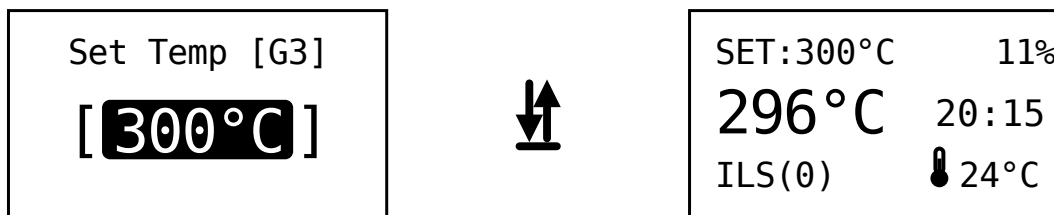
2. Turn the encoder to set the profile temperature. The temperature step is set in the menu item **[07. Stepping]**:



3. To switch to another profile, press and hold the button and turn the encoder:



4. Press the encoder button briefly: the set temperature is saved, the selected profile becomes the current one, and the station returns to operating mode:



8. Standby, Sleep and Screen Saver Modes

To save energy and extend the life of the tip and the display, the station provides several modes:

- **Standby mode** — if the station is not used for a certain time, it enters Standby mode, in which the tip cools down to the temperature set in the menu item **[02. Standby]**;
- **Sleep mode (Sleeping)** — if the station is still not in use in **Standby** mode, it switches to **Sleeping** and the tip cools down to room temperature (see menu **[03. Sleep]**);
- **Screen saver mode** — a mode in which the screen shows the current operating mode of the station and the current tip temperature, while the image moves around the display so that the OLED pixels wear evenly (see menu **[09. Scr Save]**).

Sometimes you need to switch to Standby or Sleep mode without waiting for the timer to expire. To switch from operating mode to Standby mode manually:

1. Turn the encoder counterclockwise by three or more positions — the station enters Standby mode. The temperature drops to the value set in the menu item **[02. Standby]**, which is shown in the upper-left corner of the display (for example, STB:150°C). From this moment the countdown to the automatic transition to Sleep mode begins (set in the menu item **[03. Sleep]**):



2. To switch to Sleep mode without waiting for the automatic transition, turn the encoder counterclockwise by three or more positions again while in Standby mode. The tip starts to cool slowly down to room temperature:



As soon as the tip cools below 50 °C in Sleep mode, the display switches to the date and time screen. The top line shows the date and the day of the week, the middle line shows the time, the bottom line shows the current tip temperature on the left and the handle temperature on the right (or the reading of the microcontroller sensor, see menu **[05. Cold End]**).

After the time set in the menu item **[09. Scr Save]**, screen saver mode starts:



Sleep mode (temperature < 50 °C)

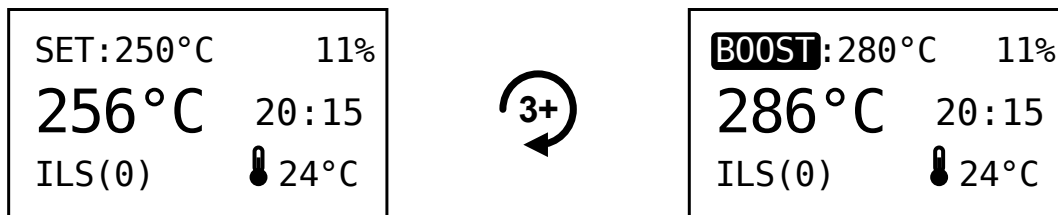
Screen saver mode

You can return to operating mode from Standby, Sleep, or screen saver mode by briefly pressing the encoder button or, depending on **Mode**, by lifting the iron and moving it a few times (see menu **[02. Standby]**).

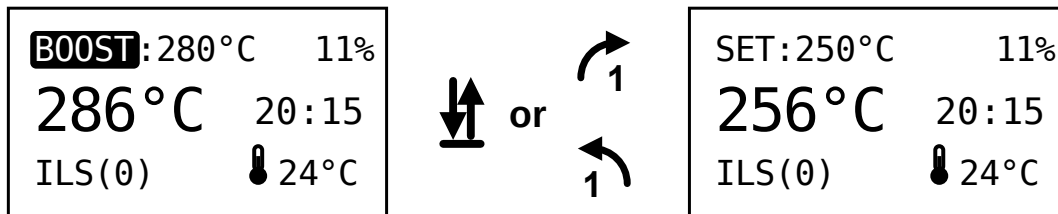
9. Boost Mode

Sometimes, while soldering, you need to briefly heat the tip above the operating temperature of the selected profile. For this purpose, the station has a boost mode (**Boost**): it quickly raises the temperature by the amount set in the menu item **[04. Boost]**. To activate **Boost** mode, do the following:

1. Turn the encoder **clockwise by three or more positions**. The tip temperature rises by the amount set in the menu item **[04. Boost]**, and the label **BOOST** and the boost temperature value appear in the upper-left corner of the display:



2. The station leaves boost mode automatically when the time set in the menu item **[04. Boost]** has elapsed, or manually — by briefly pressing the encoder button or by turning the encoder one position (in either direction):



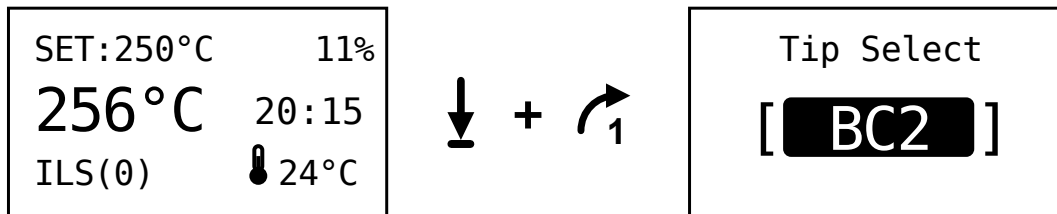
10. Tip Selection and Calibration

Tips of the T12 series have slightly different temperature characteristics across the entire operating range — even original tips of the same type. To keep the displayed temperature as close as possible to the actual temperature at the tip end, you must:

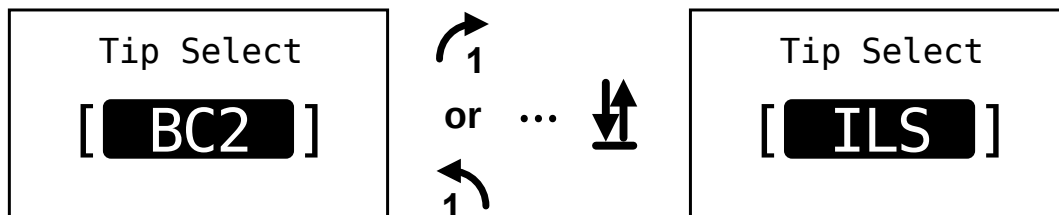
- On the **Tip Select** screen, select the tip type that is actually installed. If you select another type, the station keeps working, but holds the temperature less accurately;
- Calibrate the temperature of each tip in use at three reference points (see Section **10.2 Calibrating a Tip**);
- In the menu item **[06. Tip]**, which lists all T12-series tip types, leave enabled only those that you have or use.

10.1 Selecting a Tip

1. In operating mode, press the encoder button and, while holding it, turn the encoder clockwise — this opens the screen **Tip Select** with the list of types marked in the menu item **[06. Tip]**:



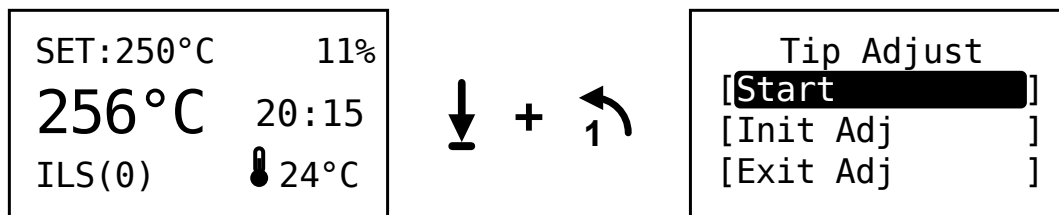
2. Turn the encoder to select the T12 tip type installed in the soldering iron and press the encoder button briefly.



10.2 Calibrating a Tip

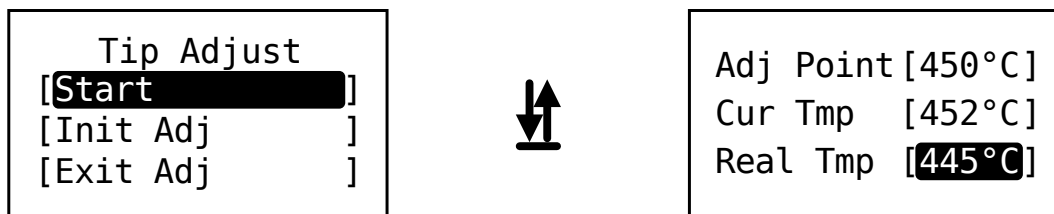
For calibration it is advisable to have a dedicated temperature meter for the soldering tip, such as the HAKKO FG-100 or similar. If no dedicated instrument is available, you can use a multimeter with a temperature measurement function. To calibrate the installed tip, follow these steps:

1. Make sure that the type of the installed tip is selected in the station (see Section **10.1 Selecting a Tip**).
2. In operating mode, press the encoder button and, while holding it, turn the encoder counterclockwise — this opens the screen **Tip Adjust**:

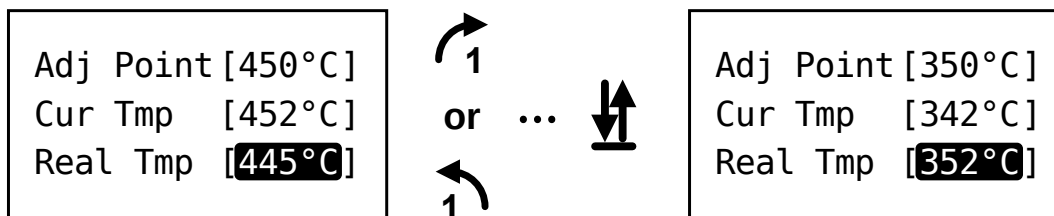


Menu items of **Tip Adjust**:

- **Start** — start calibration;
 - **Init Adj** — erase all previous calibration data for this tip type. Recommended when replacing a tip with a new one of the same type;
 - **Exit Adj** — exit the menu to operating mode.
3. Select item **Start** and press the encoder button briefly: the station moves to the first reference point. At this step the station heats the tip to approximately 450 °C (item **Adj Point**):



4. When the temperature has stabilized, measure the temperature of the tip end as accurately as possible with an external instrument. Enter the result by turning the encoder at the item **Real Tmp** and press the encoder button briefly:



5. Repeat the procedure for the second (about 350 °C) and third (about 250 °C) reference points, after which the station returns to operating mode.

For a calibrated tip, the lower-left corner of the display shows, after the tip type name, an asterisk (*) instead of (0).

11. Station Menu

To enter the menu, press and hold the encoder button for more than 2 s (see Section 4. **Selecting the Interface Language**). Menu items are selected by turning the encoder, and an item is entered with a short press of the button. To leave the settings without saving changes, press and hold the button. To exit the menu, select the item **[23. Exit]** (or press and hold the button).

Menu item **[01. PID Adjust]** contains the PID controller parameters, which **should not be changed**. Menu items **[14. Desolder]** and **[15. Pump set]** are not used with this station and are not described in this manual.

[02. Standby] — Standby Mode

Setting the tip temperature and the time before the station enters Standby mode.

Mode — sets which actions bring the station out of Standby mode. [SHAKE] — by moving the handle, pressing the encoder, or turning it one position clockwise. [SWITCH] — when the handle's motion sensor opens once. [MANUAL] — only by pressing the encoder or turning it one position clockwise. [AUTO] — by any of the SHAKE, SWITCH and MANUAL methods.

Time [10MIN] — idle time (no actions described in **Mode**) after which the station enters Standby mode.

Temp [150°C] — tip temperature in Standby mode.

Standby Setup		
Mode	[SHAKE]	←
Time	[10MIN]	
Temp	[150°C]	

[03. Sleep] — Sleep Mode

Setting the time before the station enters Sleep mode.

Dly Time [25MIN] — time spent in **Standby** mode, after which the station enters Sleep mode. The tip then cools to room temperature, and when its temperature falls below 50 °C, the date and time screen is activated.

Sleep Setup
Dly Time [25MIN]

[04. Boost] — Boost Mode

Boost mode settings.

Temp [030°C] — sets by how many degrees the temperature is raised relative to the operating temperature.

Time [05MIN] — sets the maximum duration of boost mode.

Boost Setup
Temp [030°C] ←
Time [05MIN]

[05. Cold End] — Cold Junction

Here you set the actual handle temperature, which the station takes into account when automatically regulating the tip temperature.

Mode [NTC] — select if the soldering iron handle has a thermistor. [CPU] — if there is no thermistor: in this case the sensor built into the microcontroller is used, so accuracy is lower and this option is less preferable.

Temp [25°C] — actual (measured) temperature of the soldering iron handle.

Cold End Adj
Mode [NTC] ←
Temp [25°C]

[06. Tip] — List of Tips in Use

List of T12 tip types available to the station (up to 99 entries, in alphabetical order). Leave enabled (check mark) those that you have or use, and disable the rest (cross). Then, when you change the tip, only the types you need remain in the selection list and you can quickly select the right one — the station will apply its temperature profile and the correct regulation algorithm.

```
Tip Enable
[ 01 ] [ N1-06 ] [ ☒ ]
[ 02 ] [ N1-08 ] [ ☒ ]
[ 03 ] [ N1-10 ] [ ☒ ]
```

[07. Stepping] — Temperature Step

Temperature change per encoder position when setting the tip operating temperature (see Section 7. **Setting the Temperature and User Profiles**). The value is selected from 01 °C to 50 °C. This lets you choose a convenient way to set the temperature: finer control or a wider range.

```
Temp Stepping
Stepping [ 05°C ]
```

[08. Password] — Password Lock

Enables password protection of the station and sets the time after which it locks. To unlock the station, enter the set password (4 digits, entered by turning the encoder).

Switch [OFF] — enables and disables the password lock.

LockTime [05M] — idle time after which the station locks.

Password [8888] — setting the password (4 digits).

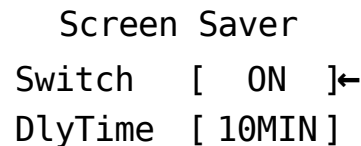
```
Password Setup
Switch   [ OFF ] ←
LockTime [ 05M ]
Password [ 8888 ]
```

[09. Scr Save] — Screen Saver

The most frequently used pixels of some OLED displays lose brightness over time, which is especially noticeable with static images. In screen saver mode the station constantly changes the picture: it moves around the display and shows the operating mode (Running, Standby, Sleeping) and the tip temperature.

Switch [ON] — enables the screen saver.

DlyTime [10MIN] — idle time after which the station enters screen saver mode.



```
Screen Saver
Switch  [  ON  ]←
DlyTime [ 10MIN ]
```

[10. Buzzer] — Audible Signals

Turns the audible signals of the station on and off. [ON] — signals on, [OFF] — “silent mode”.



```
Buzzer Setup
[  ON  ]
```

[11. Voltage] — Voltage Display

If enabled, the lower-left corner of the display in operating mode shows the supply voltage instead of the selected tip type. This is useful when the station is powered not from a stationary supply but, for example, from a battery. Values: [OFF] — disabled, [ON] — enabled.



```
Voltage Display
[  OFF  ]
```

[12. LowVol-P] — Low-Voltage Protection

Configures the behavior of the station when powered from a battery so that you notice deep discharge in time.

Function [OFF] — protection disabled.

AlarmVol — when the voltage drops to this value, the station sounds a warning signal.

Down Vol — when the source voltage drops below the set level, the station stops working.

LowVol Protect
Function [OFF] ←
AlarmVol [19.8V]
Down Vol [18.0V]

[13. Power On] — Mode After Power-On

Sets the mode of the station after power-on.

[Running] — after power-on the station enters operating mode and the tip heats to the operating temperature of the selected profile. This option is **not recommended**. [Sleep] — after power-on the station enters Sleep mode, heating is off. [Standby] — after power-on the station enters Standby mode and the tip heats to the temperature set in the menu item [02. Standby].

Power On State
[Sleep]

[16. Language] — Interface Language

Selection of the interface language: English, Chinese, Russian.

[简体中文]
[ENGLISH]
[Русский]

[17. DateTime] — Date and Time

Setting the date and time of the built-in clock. A 3 V backup battery is required for the clock to keep running when the power is off.

Date & Time

2026-09-23

20:15:10

[18. RTC Adj] — Clock Rate Correction

Fine adjustment of the clock accuracy.

Day Error — enter the daily correction in seconds (with sign, for example +00S). To determine the correction more precisely, set the clock against a time signal, compare it with the same source ten days later at the same hour, divide the resulting error by ten, and enter the result in this item.

RTC Adjust

Day Error[+00S]

[19. RTC Init] — Clock Reset

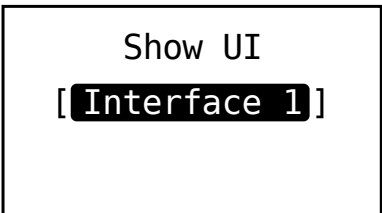
Returns the clock of the station to its initial settings.

[Confirm]

[Cancel]

[20. Show UI] — Interface Variant

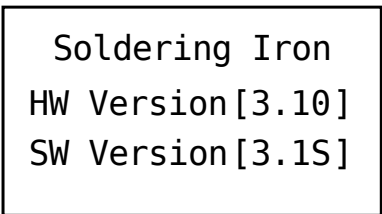
Selection of one of two variants of the main screen: with the full set of information, or with less information but larger digits (for example, of the temperature).



Show UI
[**Interface 1**]

[21. Sys Info] — Station Information

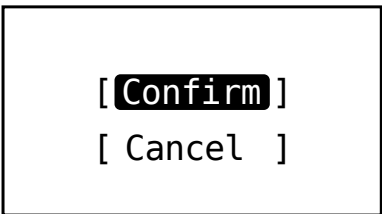
Shows the hardware version (**HW Version**) and the firmware version (**SW Version**) of the station.



Soldering Iron
HW Version[3.10]
SW Version[3.1S]

[22. Init] — Reset Settings

Returns all station parameters to factory values. After this item is selected, you will be asked to choose the interface language. You must then configure all menu items again and repeat the calibration of each T12 tip.



[**Confirm**]
[Cancel]

[23. Exit] — Exit

Exits the menu and returns to operating mode.

12. Quick Control Reference

Action	Result
Short press (in operating mode)	Setting the temperature and selecting profile G1–G5
Press and hold, turn one position (on the screen Set Temp)	Profile switching: clockwise — next, counterclockwise — previous
Turn counterclockwise by three or more positions	Standby mode (Standby); from Standby again — Sleep mode (Sleeping).
Turn clockwise by three or more positions	Boost mode (Boost).
Press and hold, turn clockwise	Tip selection (Tip Select)
Press and hold, turn counterclockwise	Tip calibration (Tip Adjust)
Long press (more than 2 s)	Enter the station menu (or exit the menu). Exit the menu settings without saving.
Short press in Standby , Sleeping and screen saver modes	Return to operating mode